

Documentation

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Preface

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1 Clinical Core Methods

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1.1 Background

This document provides information about the ABC-DS study and documentation of data currently available on the LONI website. The initial ABC-DS study began in 2015. Funded by the National Institutes on Health (NIH) through the **U01** mechanism, from 2015-2020, ABC-DS was the combined/harmonized effort of two groups of investigators: 1) the Alzheimer's Dementia in Down Syndrome (**ADDS**) group (Columbia University/NYS Institute for Basic Research in Developmental Disabilities, Massachusetts General Hospital, University of California Irvine); 2) the Neurodegeneration in Aging Down Syndrome (**NiAD**) group (University of Pittsburgh, University of Wisconsin Madison, University of Cambridge, Banner Institute, Washington University).

From 2020-2026, the ABC-DS study was funded by NIH through the **U19** mechanism. The number of participating institutions increased to ten (Columbia University/NYS Institute for Basic Research in Developmental Disabilities, Massachusetts General Hospital, University of California Irvine, University of Pittsburgh, University of Wisconsin Madison, University of Cambridge, Washington University, Kansas University Medical Center, and the University of Puerto Rico). A single, harmonized protocol, including many of the ADDS and NiAD measures, was adopted by all ten institutions. The U19 project includes new participants as well as individuals continuing from the U01 projects.

While the focus of ABC-DS is on individuals with Down Syndrome, NiAD also recruited sibling controls, and unaffected siblings continue to be recruited in the U19. In the ongoing U19 study, in addition to new participants, as many former U01 participants as possible have been recruited, for continued evaluation.

ABC-DS participants are scheduled to be seen every 16 months. Sibling controls receive a shorter cognitive evaluation battery than participants with Down Syndrome and do not receive caregiver assessments.

1.2 Available Data

ABC-DS data has been harmonized and made available to the research community through the Laboratory of Neuroimaging website (LONI). Cumulative data released at annual intervals include evaluations in which a consensus diagnosis has been generated as of a set date, based on a defined battery of tests. Currently, data on a total of 1,274 evaluations conducted on 503 participants with Down Syndrome (cases) are available on LONI; on 503 individuals with Down Syndrome and 59 sibling controls, along with 161 evaluations conducted on 59 sibling controls. Case and control data are provided in separate tables.

Data collected at multiple evaluations are organized in a one-record-per-evaluation format sorted by participant ID, subject label, and consecutive evaluation (event_sequence), regardless of study of recruitment. For example, a participant with three evaluations in the U01 and one evaluation in the U19 would have data for event_sequence 1, 2, 3 and 4 (the first U19 evaluation). A participant seen in the first and third U01 evaluation with one evaluation in the U19 would have data for event_sequence 1,2 and 3. Note that event_sequence is numbered consecutively, regardless of whether a particular 16 month month evaluation (in this case the second U01 evaluation) is missed. The age_at_event_and_latency table provides the age of the participant at each evaluation and the latency in days from baseline of each subsequent evaluation. The number of evaluations for which data are currently available is shown in Table 1. Please note that the N of evaluations available for any specific measure is affected by a number of factors (please see descriptions of individual measures, below).

Table 1. Number of available ABC-DS evaluations

Evaluation (event_sequence)	Down Syndrome Cases	Sibling Controls
1st (baseline)	503	59
2nd evaluation	346	35
3rd evaluation	283	17
4th evaluation	138	5
5th evaluation	4	1
Total evaluations	1,274	116

1.3 ABC-DS Cognitive & Caregiver Measures (2015-2026)

Some measures in the harmonized data available on LONI were collected in the U19 study, but not in one or both of the U01 studies (ADDS and/or NiAD); for the tables containing these measures (flagged in the descriptions below), sample sizes will be smaller than for measures administered in all three studies.

1.4 Measures in the ABC-DS Integrated Dataset

1.4.1 Cognitive Assessment Measures

For all the cognitive measures, before performing any modelling, please check the distributions of the variables. The vast majority of them have a skewed distributions and are zero inflated, which requires either some type of transformation or proper modelling.

- a) The **Down Syndrome Mental Status Examination (DSMSE)** (an enhanced version)(Haxby 1989) provides an omnibus measure of neuropsychological functioning, assessing recall of personal information, orientation, immediate and delayed memory, language, visuospatial function and praxis.
- b) The Rapid Assessment for Developmental Disabilities (RADD)(McCarthy 1972) Assesses the ability of the individual to copy simple hand movements. For ABC-DS we are only using the Hand Movements and Imitation Subtest of the measure.²
- c) The McCarthy Scales of Children's Abilities: Category Fluency Test(Korkman et al. 2007) An adaptation of the McCarthy Category Fluency Test has been employed to assess changes in language ability and frontal lobe function over time. It requires participants to name as many foods and animals as they can within 20 seconds. We have been using a single summary score generated by adding the number of correct responses for each category.²
- d) The NEPSY (2nd ed.) Word Generation Semantic Fluency Subtest(Korkman et al. 2007) assesses verbal productivity through the ability to generate as many words as possible in 20 sec. within a specific semantic category (e.g., animals, food and drinks). The score is the total number of unique animal names/foods/drinks generated.²
- e) Kaufman Brief Intelligence Test (2nd ed)(Kaufman 2004) The KBIT-2 is a brief measure of verbal and nonverbal intelligence for ages 4-90. It provides crystallized (verbal) and Fluid (nonverbal) score along with an IQ composite score.¹
- f) The Wechsler Intelligence Scale for Children, 4th Edition (WISC-IV)(Wechsler 2003) Block Design subtest was used to measure visuo-spatial organization. The subtest is a core perceptual reasoning timed test that requires that the individual reproduce designs using blocks that have red surfaces, white surfaces, and surfaces divided diagonally into half red and half white. The Digit span forward subtest of the WISC-IV was used to assess rote learning and memory. The WISC-IV version is somewhat easier (as it is designed for children ages 6 to 16). It contains series ranging in length from two to nine digits and primarily involves rote learning and memory. Digit forward requires participants to recite a series of digits in the exact order given by the evaluator (forward). Conversely, Digit Span backwards requires participants to recite series of digits backwards and taps ability to manipulate verbal material in working memory.

- g) Haxby Extension.(Haxby 1989) Like the WISC Block Design subtest, the Haxby Extension requires that participants reproduce designs using blocks with red surfaces, white surface, and surfaced divided diagonally into half red and half white. However, the designs are simpler than those on the WISC, such as starting with reproducing a single block. Possible cores range from 0-16.
- h) The Beery-Buktenica Developmental Test of Visual Motor Integration (VMI)(Beery 2004) involves the copying of figures of progressive complexity using paper and pencil. It is designed to assess the extent to which the individual can integrate his/her visual and motor abilities. Scores range from 0 to 30 (higher scores indicating more correctly copied items).
- i) The Modified Cued Recall Test (mCRT)(Devenny et al. 2002) is a list-learning task that measure verbal learning and memory. The mCRT consists of a training period in which 12 items are presented, 4 at a time, with each item accompanied by a unique category cue. The testing phase consists of 3 trials which generates a free recall score (FRS -spontaneous recall of the 12 items), cued recall score (CRS-items only recalled when the category cue is provided), a total score (sum FRS and CRS), and an intrusion score (items named that were not on the original list).
- j) The Modified Cats and Dogs Task(Ball et al. 2008) is an adapted version of the Stroop Task and it assesses cognitive inhibition. Specifically, the task measures an individual's ability to inhibit a natural response (naming pictures of a dog as "dog") and to replace it with an incongruent one (naming pictures of a dog as "cat").
- k) The Purdue Pegboard Test(Tiffin and Asher 1948) measures manual dexterity (left and right hand separately) as a participant places pins/pegs in a pegboard with 2 parallel rows.
- l) The Cancellation Task-Revised(Krinsky-McHale et al. 2008) measures processing speed, visual selective attention, vigilance, perceptual speed and visuomotor ability by asking participants to circle a target stimulus/picture on a page containing both target pictures and distractor pictures.
- m) The Tinetti Assessment Tool: Gait subtest(Tinetti 1986) quantifies gait in 7 domains by observing participants while walking. Those domains are: Initiation of Gait, Step Length, Step Height/ Foot Clearance, Step Symmetry, Step Continuity, Path, Trunk and Walking Stance.

Caregiver Interview Measures

- a) The Dementia Questionnaire for People with Learning Disabilities (DLD).(Eurlings et al. 2006) The DLD consists of 50 items in 8 subscales, including short-term memory, long-term memory spatial and temporal orientation, speech, practical skills, mood, activity and interest, and behavioral disturbances. The DLD generates two scores, one reflecting cognitive abilities (DLD/SCS) and one reflecting social skills (DLD/SOS).

- b) The NTG Early Detection Screen for Dementia (NTG-EDSD).(Esralew et al. 2017)
The NTG-EDSD includes sections on behavioral/cognitive domains that could reflect indications of dementia clinical progression. In the current ABC-DS cohort, the DLD and NTG significantly differentiated participants who were Unaffected from those with MCI-DS and Dementia.
- c) The Vineland Adaptive Behavior Scales, 3rd edition (VABS-3),(Sparrow et al. 2016)
Domain Level Parent/ Caregiver Form, provides a standardized assessment of functional abilities for use with individuals (ages 0 – 90+) having intellectual and developmental disabilities. The Domain Level Parent/Caregiver Form is administered as a semi-structured interview with the parent or caregiver and focuses on home and family- life behavior. Adaptive functioning within three domains, Communication, Daily Living Skills, and Socialization, are assessed.^{1 2}
- d) The Reiss Screen for Maladaptive Behavior(Reiss 1994) was created specifically for use with individuals with ID to screen for possible depression, psychosis and behavior problems. Caregivers are asked to rate 38 symptoms of psychiatric behavior as being “No Problem”, “Problem”, or “Major Problem” for the individual in question. Behaviors are categorized into 8 sub-domains reflecting (a) aggressive behavior, autistic-like behavior, (c) psychosis, (d) paranoia, (e)/(f) behavioral and physical symptoms of depression, (g) dependent personality disorder, and (h) avoidant disorder.
- e) The Neuropsychiatric Inventory (NPI)(Cummings et al. 1994) is a validated instrument used to assess neuropsychiatric symptoms in dementia. The questionnaire asks caregivers to retrospectively rate symptoms in 12 neuropsychiatric domains: delusions, hallucinations, agitation/ aggression, dysphoria/depression, anxiety, euphoria/elation, apathy/indifference, disinhibition, irritability/lability, aberrant motor behaviors, nighttime behavioral disturbances, and appetite/ eating disturbance.²

Health Status and Life Events

- a) Demographics Questionnaire; includes year of participant’s birth; sex; ethnicity and race; primary language; age of mother and of father at participant’s birth; handedness; primary residence; level of independence; activities (e.g., vocational training program, working in the community, postsecondary education program).
- b) Medical Health History Questionnaire; includes Psychiatric conditions (e.g., depressive disorder, anxiety disorder); Neurological conditions (e.g., dementia, stroke, seizure disorder); Eyes/ears/nose/throat (e.g., cataracts, visual impairment, hearing impairment); Cardiovascular conditions (e.g., congenital heart disease, pacemaker); Respiratory conditions (e.g., asthma, pneumonia, sleep apnea); Allergies; Dermatologic/Connective tissue conditions (e.g., alopecia, rosacea); Musculoskeletal conditions (e.g., osteoporosis; gout); Endocrine/metabolic conditions (e.g., diabetes, Vitamin B12 deficiency); Menstrual information (e.g., menopausal); Hormone replacement information; Gastrointestinal conditions

(e.g., celiac disease, irritable bowel syndrome); Hematopoietic conditions (e.g., anemia); Cancer; Surgery; Covid-19; and Other conditions;

- c) **Stress Index:** Caregivers were asked about the presence or absence of significant life events and were asked to rate how the participant handled the experience.²

1.5 Special Cautions When Using this Dataset

Modified Cued Recall Test. Two sets of stimuli images (Set A/ Version 1 or Set B/ Version 2) were used. The set used varied across site and data collection cycle. Analyses conducted thus far suggest that set/version administered influences participant performance. The following variables are unique to the ABC-DS U19 study: `crset`, `ift1r`, `ift2r`, `ift3r`, `tifrrsps`, `agfrin1`, `agfrin2`, `agfrin3`, `tafi`, `c1t1`, `c1t2`, `c1t3`, `c2t1`, `c2t2`, `c2t3`, `c3t1`, `c3t2`, `c3t3`, `crc1`, `crc2` and `crc3`.

Down Syndrome Mental Status Examination (DSMSE): A component of the DSMSE, the Shoebox Memory Test, was unique to the ADDS U01 and ABC-DS U19 studies; thus, sample sizes for the `colshbx` and `dsto2` will be smaller than those for other DSMSE variables. Similarly, variables `colshbxtime`, `dsmeltime`, `colmemscr` and `nonmemory` are unique to the ABC-DS U19 study; thus, sample sizes for these variables will be smaller than those for other variables.

Modified Cats and Dogs Test: The variables, `nttrain` and `strain`, are new and unique to the ABC-DS U19 study.

NTG-Early Detection Screen for Dementia (NTG-EDSD): The Caregiver Concern Questions, variables `ntspcon` and `ntspconage`, are unique to the ADDS U01 and ABC-DS U19 studies.

Purdue Pegboard Test: Variables `pp_dh_tr` and `pp_ndh_tr` are unique to the ABC-DS U19 study. The variable `de_handedness` can be used to determine participants' dominant hand (1 = Right-handed, 2 = Left-handed, 3 = Ambidextrous).

Stress Index: The variable `si_victim` is a new variable unique to the ABC-DS U19 study.

Tinetti Gait Assessment: The following variables are unique to the ABC-DS U19 study: `tbagait`, `tbaslh`, `tbafe`, `tbass`, `tbasce`, `tbap`, `tbat`, and `tbawt`.

Verbal Fluency- McCarthy Category Fluency Test. The "Exceptional Scoring Method" is used. Variables `vffdexpr` and `vfanexpr` are unique to the ADDS U01 and ABC-DS U19 studies; the variable `vftotexpr` is unique to the ABC-DS U19 study.

1.6 Protocol for Sibling Controls

Approximately 50 sibling controls were recruited and underwent the same procedures with the exception of no caregiver assessments and a small cognitive screening battery. In addition, during the 2015-2020 U01 grant, only baseline blood draws were conducted. Sibling controls provided medical histories and were screened using the AD8(Galvin et al. 2005) (a brief informant interview to detect dementia) and the Montreal Cognitive Assessment (MoCA).(Nasreddine et al. 2005) The two screens were used to simply ensure that the siblings did not have MCI/Dementia and are not included in the ABC-DS database.

2 Genomic Data

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GENOMIC DATA (GENOME-WIDE SNP MICROARRAY)

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Summary. This dataset includes genome-wide microarray data for 373 adults with Down Syndrome (DS) and 43 sibling non-DS controls (total N=416) in the Alzheimer’s Biomarkers Consortium — Down Syndrome (ABC-DS) study. This genomic data set includes data on study participants who were enrolled in the U01 part of the ABC-DS study as of 4/30/2022.

2.1 Genotyping

Genotyping was performed at the Center for Applied Genomics at Children’s Hospital of Pennsylvania using the Illumina Infinium General Screening Array (GSA) Version 2, with added disease markers for all samples, except for 25 participants for which Version 3 was used. In total, 759,993 and 654,027 variants across the genome were genotyped on GSAv2 and GSAv3, respectively. The two versions of the genotyping microarray had 619,856 (~95%) overlapping variants. We have provided several subsets of the complete data set (see “Summary of the data files provided” for more details). Variants in these data were mapped to the GRCh37/hg19 assembly.

2.2 Trisomic Chromosome 21

Chromosome 21 was treated separately because traditional tools cannot be applied to trisomic chromosome 21. Trisomic variants were treated as copy number variants (CNV) and were called in using the cnvPartition CNV Analysis (v3.2.0) plug-in in GenomeStudio 2.0 with Genotyping module.(Illumina, Inc. 2020a, 2020b) Allele specific genotypes were then exported using the CNV Region Report plug-in (v.2.1.2)(Illumina, Inc. 2020b). In total, 9,785 and 8,446 variants were genotyped on GSAv2 and GSAv3, respectively. As done commonly, the p-arm of Chr. 21 was dropped from analysis because it is known to be highly heterochromatic with numerous repeat sequences. In the current data, this characteristic resulted in high rate of variants called as disomic in among those karyotyped with DS and discrepancies between duplicates in this region. For these reasons, all variants genotyped on the p-arm should be excluded from analysis.

2.3 Genotype quality control

To maximize the flexibility of these data for all investigators, we aimed to minimize the likelihood of distributing erroneous data, while minimizing the quality control steps. For example, we included SNPs that did not achieve high genotyping rates, which would have been dropped in a standard analysis practice. Specifically, we checked for sex-mismatch cases by identifying cases where sex obtained from the analysis of X-chromosome SNPs using PLINK(Chang et al. 2015) did not match the reported sex. These cases were re-confirmed with research coordinators to rule out potential data entry error, etc. Relationships (e.g., siblings, monozygotic twins) among subjects were confirmed by pair-wise genome-wide estimates of proportion identity-by-descent (IBD) computed using the KING(Manichaikul et al. 2010) software. Discrepant relationships were reviewed against available demographic and family data (when siblings are genotyped) to determine the most likely relationship consistent with the proportion of IBD or other scenarios (i.e., sample duplicates, duplicate enrollments across cohorts). Samples with unresolved or ambiguous identities arising from mismatched sex and discrepant relationships were excluded from the dataset. In duplicated samples, the samples with the lower genotyping call rate were dropped from the data set. In the end, data were provided for 416 participants.

2.4 Imputation

We are not providing imputed data and suggest the TOPMed Imputation Server for genotype imputation.

2.5 Summary of the data files

Nomenclature Subject Identification (ID) Number

- For non-DS control siblings, the Individual ID (IID) of the DS-proband was used as the Family ID (FID)
- For families consisting of multiple participants with DS, the lower valued IID was used as the FID
- In data files related to Chromosome 21, participant IDs were formatted as follows: “FID_IID”

Trisomic Chromosome 21 genotypes

- **ABC-DS.LONI.Chr21.GRCh37.CNCounts.txt**: Copy number count at each genotyped position.
- **ABC-DS.LONI.Chr21.GRCh37.Dosage.txt**: Dosage of the “A” allele (“A_Allele” column). Dosage can be 0, 1, 2, or 3 copies.
- **ABC-DS.LONI.Chr21.GRCh37.Genotype.txt**: Genotype in ACGT coding

Non-Chromosome 21 genotypes: These genotypes are provided in binary PLINK data files (i.e., bed/bim/fam files) with pedigree and sex information.

PLINK files consisting of *all variants* genotyped (i.e., the union of GSAv2 and GSAv3 chips)

- **ABC-DS.GRCh37.non-Chr21.LONI.allSNPs.bed**
- **ABC-DS.GRCh37.non-Chr21.LONI.allSNPs.bim**
- **ABC-DS.GRCh37.non-Chr21.LONI.allSNPs.fam**

PLINK files consisting of *only overlapping* variants (i.e., the intersection) between GSAv2 and GSAv3 chips

- **ABC-DS.GRCh37.non-Chr21.LONI.bed**
- **ABC-DS.GRCh37.non-Chr21.LONI.bim**
- **ABC-DS.GRCh37.non-Chr21.LONI.fam**

PLINK files consisting of the *remaining non-overlapping* variants

- **ABC-DS.GRCh37.non-Chr21.LONI.nonOverlappingVars.bed**
- **ABC-DS.GRCh37.non-Chr21.LONI.nonOverlappingVars.bim**
- **ABC-DS.GRCh37.non-Chr21.LONI.nonOverlappingVars.fam**

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3 Plasma NT1-tau in ABC-DS

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Summary (or Abstract). Our group recently developed a high-sensitivity plasma immunoassay for a novel Nterminal tau fragment termed NT1-tau on the Quanterix Simoa HD-1 platform.(Chhatwal et al. 2020; Chen et al. 2019; Mengel, Janelidze, et al. 2020; Mengel, Liu, et al. 2020) This assay succeeded at discriminating AD and mild cognitive impairment due to AD from normal controls. We have recently moved this assay to the Quanterix HD-X platform and obtained NT1-tau values for ABC-DS Wave 1 plasmas, with plans to analyze its predictive power for onset of cognitive decline in people with Down syndrome.

3.1 Method

Consumables and reagents other than antibodies were obtained from Quanterix (Lexington, MA). The tau capture antibody BT2 (194–198) was conjugated onto paramagnetic beads at 1.8 mg/mL. Detector antibody Tau12 was biotinylated according to the manufacturer using a ratio of 40 parts biotinylation reagent to 1-part antibody. Plasmas were centrifuged at $14,000 \times g$ for 4 min and then diluted 1:4 with Tau 2.0 sample diluent reagent (Quanterix). Tau210 untagged standard was diluted linearly with Tau 2.0 sample diluent to a concentration range of 270–0.02 pg/mL. Sample, standards and blanks were prepared in 1.5 mL low-binding Eppendorf tubes and were analyzed in triplicate.

The NT1 Simoa assay utilized a 3-step protocol and was performed at ambient temperature on a Simoa HD-X analyzer (Quanterix Corp.). In step 1, 100 μ l of standard, blank, or sample were added to beads coated with capture antibody and mixed for 30 min. The beads were then

harvested and washed with wash buffer. In step 2, biotinylated detection antibody (0.6 g/ml) was added and incubated for 10 min 30 s, and the beads were then washed three times. In step 3, 150 pM streptavidin- α -galactosidase was added, and following a further wash step, enzyme substrate (resorufin α -D-galactopyranoside) was added. The bead-bearing complexes were then resuspended and loaded into Simoa arrays, each containing 216,000 femtoliter-sized wells. The average enzyme unit per bead (AEB) was determined as described previously.(Mengel, Janelidze, et al. 2020) Standard curves of AEB vs. Tau210 concentration were fitted to a four-parameter logistic function with 1/Y2 weighting.

The samples were analyzed over five runs spanning eight days, with pooled-plasma internal standards run on each plate to ensure reproducibility across runs. The Simoa HD-X instrument caused plate wells to be randomly dropped from the analysis, resulting in 13% of samples being analyzed with two instead of three technical replicates, and a single sample (0.3%) to be analyzed in single replicate. Detailed data on internal controls, coefficients of variation, and dropped values or any other technical aspects are available upon request.

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4 Plasma P-tau217

Plasma tau phosphorylated at threonine 217 (P-tau217) and glial fibrillary acidic protein (GFAP) assays

Last Revised: October 20, 2021

VERSION 1.0

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Summary (or Abstract). Plasma concentrations of tau phosphorylated at threonine 217 (P-tau217) was measured using immunoassay developed by Lilly Research Laboratories. Plasma glial fibrillary acidic protein GFAP concentration was quantified using Simoa kit (Quanterix, Lexington, MA, USA)

4.1 Method

Plasma P-tau217 concentration was measured according to the published protocols using immunoassay on a Mesoscale Discovery platform developed by Lilly Research Laboratories.^{1,2} Briefly, biotinylated-IBA493 was used as a capture antibody and SULFO-TAG-4G10-E2 (anti-Tau) as the detector and samples were diluted 1:2. The assay was calibrated with a synthetic Ptau217 peptide. Plasma GFAP concentration was quantified using Simoa kit (Quanterix, Lexington, MA, USA) according to the manufacturer's instructions. One sample with insufficient plasma volume was only analyzed with the P-tau217 assays. All measurements were above the lower limit of detection of the assay (P-tau217, 0.10 pg/ml; GFAP, 0.26 pg/ml). The inter-assay coefficient of variation (CV) for 3-4 quality control samples included in every run were 14.2% and 17.4% in the P-tau217 and GFAP assays, respectively. The intra-assay CV was 5.7% in the P-tau217 assay and 4.0% in the GFAP assay. All samples were analyzed by staff blinded to the clinical and imaging data.

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5 Proteomics

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5.1 Background

Serum and plasma samples were processed and stored at the University of North Texas Health Science Center Institute for Translational Research and were run across three waves (Wave 1: 2019; Wave 2: 2021; Wave 3: 2022). Blood samples were initially prepared through an automated system utilizing a customized Hamilton Robotics StarPlus system. Proteomic assay kits were purchased from select vendors (Meso Scale Discovery, Quanterix). Samples were run in singlicate, with a subset run in duplicate. Control samples were included across plates and runs for quality control.

5.2 Methods

Electrochemiluminescence (ECL)

Proteomic assay kits were commercially obtained from Meso Scale Discovery (MSD; <http://www.mesoscale.com>). Per our previously published methods,(M. Petersen et al. 2020; M. E. Petersen et al. 2020) samples were assayed using electrochemiluminescence (ECL) technology. The ECL platform has been extensively applied to assay biomarkers associated with the range of neurodegenerative diseases including Alzheimer’s disease (AD). The UNTHSC laboratory maintains a database of over n>2000 samples from the neurotypical AD population conducted in duplicate using the ECL technology on the referenced MSD platform with average coefficient of variation (CVs) < 10%. A total of 500ul of plasma and serum were assayed for the following proteomic markers: fatty acid binding protein 3 (FABP3), beta 2 microglobulin (B2M), Factor VII, I-309, pancreatic polypeptide (PPY), c-reactive protein (CRP), thrombopoietin (TPO), interleukin (IL)-5, IL-6, IL-8, IL-10, IL-18, alpha 2 macroglobulin (A2M), eotaxin 3, vascular cell adhesion molecule-1 (sVCAM-1), soluble intercellular adhesion molecule-1 (sICAM-1), thymus and activation regulated chemokine (TARC), tenascin C, tumor necrosis factor-alpha (TNF-a), and serum amyloid A (SAA) (see Table 1). Five pooled plasma and serum control samples were included on each MSD plate

layout reflecting a similar distribution of age for Down Syndrome (DS) samples and sibling controls.

Single Molecular Array (SIMOA)

Proteomic assay kits were commercially obtained from Quanterix (<http://quanterix.com>). A total of 500 ul of plasma was assayed using Single Molecule Array (SIMOA) technology using HD-1 imager for some of Wave 1 assays and HD-X imager for Wave 1, Wave 2, and 3 (see Table 2). The following proteomic markers were derived: Amyloid beta (A) 40 and 42, total tau, ptau181, glial fibrillary acidic protein (GFAP) and neurofilament light chain (NfL). Five pooled plasma control samples were included on each SIMOA plate layout reflecting a similar distribution of age for DS samples and sibling controls.

5.3 QA/QC

All supplies for key biological resources were purchased from multiple vendors via commercially available kits (e.g. Meso Scale Discovery; Quanterix). Each of these vendors conducts internal validation of resources. Protocols are conducted to ensure reproducibility of findings. Protocols include validation of resources, which spans monitoring dilution linearity, spiked recovery, monitoring of low and high end of detection ranges to avoid lack of reliability at restricted ranges, examination of matrix effects, inclusion of pooled standards across lots, examination of inter- and intra-assay variability, real-time monitoring of coefficient of variation (CVs) and monitoring of all lot numbers across experiments.

Table 1. Lot Number and Assay Information (Meso Scale Discovery)

Wave: Wave 1 Wave 2 Wave 3

Sample: Plasma Serum

Table 2. Lot Number and Assay Information (Quanterix)

Wave: Wave 1 Wave 2 Wave 3 Wave 4

Error Coding: -777777 Missing; -999999 LLDL

5.4 Updates to Previous Release

Column Name

Date Effective

Reason/Information

Action

Project

4/25/2025

Internal identifier

Removed from dataset

BatchName

4/25/2025

Internal identifier

Removed from dataset

Conc_Merged

4/25/2025

Combined into a single column. From wave 4 onwards, use Comment_30%CVflag to identify if singlet or duplicate data.

Removed from dataset

Comment_30%CVflag

4/25/2025

Information on interpretation is available in the cell comments.

Removed from dataset

Comment_Instrument

4/25/2025

Data error from instrument

Removed from dataset

6 Frequently Asked Questions

Last Revised: October 10, 2024
VERSION 1.0

6.1 Overview

6.1.1 Database Structure

What is the overall structure of the database?

The research design calls for evaluating each subject at intervals of approximately 16 months. The data available on LONI are organized into separate tables, each representing and named for a particular data domain, e.g. Neuropsychiatric_Inventory, Registration, Premorbid Functioning. For data collected at each evaluation, tables are organized in a one-record-per-evaluation format, sorted by ID (subject_label) and consecutive evaluation number (event_sequence). The latency in days between the baseline evaluation and each subsequent evaluation, and the age of the subject at each evaluation are documented in the table Age_at_Event_and Latency. Data types collected only once for each subject have one record per subject_label. Tables contain data collected from control subjects have the suffix ‘_Control’ in the table name. All other data tables contain data collected from subjects with Down Syndrome.

6.1.2 Data Tables on LONI

How are data tables organized on the LONI website?

As shown in the table below, data are organized by domain. After selecting a domain, all tables in that category are shown; individual tables or all tables may be selected and downloaded.

6.1.3 Documentation

Is there documentation for each domain, in addition to information provided in this FAQ document?

Yes, a document with information specific to each domain has provided by project leaders responsible for the collection and processing of each data type. These documents appear in each of the domain-specific tabs on the LONI website.

6.1.4 Data Sources

What is the source of available data?

The data available on LONI come from three different projects: ADDS and NiAD, funded through a U01 mechanism, and the ongoing ABC-DS U19 project. The available data have been harmonized across all three projects. ADDS and NiAD collected non-overlapping samples of Down Syndrome cases. The current U19 project includes subjects continuing from the U01 projects as well as newly recruited subjects, i.e. individuals not previously seen in ADDS or NiAD. ADDS did not collect data on controls, while NiAD and the current U19 project recruited sibling controls of Down Syndrome subjects.

6.2 Data Dictionary

6.2.1 Dictionary and Codebook

Is a data dictionary available?

Yes, a data dictionary containing the names and descriptions is available. For categorical variables, labels of each numeric value for each variable are provided in a separate codebook.

6.2.2 Free Text Columns

In the data dictionary, what are the field_types “D” “N” and “T”?

There are free text columns in the EDC, but they are omitted from the external investigator-facing data freezes.

6.2.3 Event Sequence

What does ‘event sequence’ refer to?

Event_sequence is just that, an indication of the sequential number of times the subject has had across the ABC-DS study.

6.2.4 Seizure Variable

Where is the ‘seizure’ variable in the harmonized dataset?

We no longer administer the health questions on the NTG-EDSD form in the U19, so the seizure variable wouldn’t be in the harmonized data set (U01/U19). However, there is a variable in the health history form (hh_seizure). This will only tell us if there has ever been a seizure diagnosis. It does not tell us whether it is currently an issue. We had to collapse the Recent/Active and Remote/Inactive variables into just yes/no in the harmonized data set.

6.3 Cognitive and Caregiver Measures

6.3.1 Cognitive Inventory

Is there an inventory of which neuro/psych/cognitive tests are available for ABC-DS participants?

Yes, you will find these in the Clinical Core Methods Document posted on LONI or available [here](#)

6.3.2 Vineland Data

There is only one assessment per subject and the `event_sequence` takes different values depending on the subject. Does this imply this assessment was only collected once per subject and possibly at different visits (not necessarily at baseline)?

Each participant only has one Vineland record. It might have been done in a visit that was not the Baseline visit.

6.3.3 ABC-DS IQ Data

The variables MTAGE1, MTAGE2, MTAGE3 in the ABC-DS IQ Worksheet data do not seem to be included in the documentation provided. Q: Can you please point to any data specifications where these variables are described.

Not many participants had alternate additional sources. “MTAGE” stands for Mental Age for source 1, 2, and 3.

6.3.4 Cognitive Distributions

Are all the distributions normal for the cognitive tests?

For all the cognitive measures, before performing any modelling, please check the variables distributions. The vast majority of them have a skewed distributions and are zero inflated, which requires either some type of transformation or proper modelling.

6.4 Adverse Events

Do you know if we can find what adverse events were recorded?

The only AE data in the harmonized dataset is cause of death if occurred.

6.5 NeuroImaging

6.5.1 MRI Scans

Is it possible to see MRI reports or a list with the # and type of abnormal findings, e.g., areas of siderosis or micro-/macro-hemorrhages?

This data is not available in the harmonized dataset.

6.5.2 Image Processing

I was not able to view or add post-processed images to my data collection and also wondered what kind of pre-processing and post-processing was done.

Our Neuroimaging Core leads are Brad Christian and Adam Brickman. You can contact them for more information. In the future there will be a Methods document available on LONI for Neuroimaging.

6.5.3 Structural MRI Scans

Can you provide any information on decisions about which T1 sequences were used at which sites / visits. The dataset contains several variations of MPRAGE and SPGR sequences, and we want to ensure that our analyses are not biased in any way by the different types of sequences.

The specific acquisition parameters should be contained within the header information for each scan. For the U19 data, T1-weighted scans were collected using the ADNI 4 protocol, set up by the Mayo group for the different scanners (I think most are on Siemens, maybe one or two sites are on GE). Prior to the U19, there were two separate consortia (we sometimes call this the “U01 data”), including NiAD & ADDS. The T1-weighted scan parameters were harmonized within consortium but not necessarily across consortia.

6.5.4 Available Imaging Results

What analysis results data is available on LONI?

The only analysis results currently at LONI are those for centiloid for the former NiAD/ADDS study.

6.5.5 Centiloid

How does the centiloid file interact with the `event_sequence`?

Currently it does not include the `event_sequence` and it is all “U01” data. In the future the `event_sequence` will be added.

6.6 Clinically Significant Findings

Does ABCDS-U19 ClinSigFindings 2 include all abnormal MRI findings from the U19 cohort?

Yes, it contains all the findings that were reported and entered into the EDC by the time of the data freeze Health History and Concurrent Medications.

6.7 Health History and Medications

Is there any information on concurrent medications and prior medical history in Freeze 4 data?

Yes, these tables are available on LONI.

6.8 Lookup Table

Is there a look up table for all these IDs?

Yes, the ‘crosswalk’ is in the internal analytic file, available to ABC-DS internal investigators

Is there an indicator of whether or not a participant is on TRC-DS since treatments can alter biomarker results?

Yes, we know if they were/are co-enrolled, but that doesn’t mean they are in a clinical trial (yet); the eCRF/data for clinical trial participation is in progress as of 9/2024.

6.9 Harmonization

Is imaging and IQ data harmonized between ADDS and NiAD?

For imaging, it depends on the measure; amyloid PET is harmonized using centiloids; MRI and Tau PET are not harmonized. If amyloid PET is used for analyses then centiloids should be used since centiloids are a standardized measure of amyloid deposition. Site should still be included in all analysis as a covariate regardless of harmonized data or not. All derived MRI measures (cortical thickness and volumetrics) as well as Tau PET are not harmonized and at least Combat should be employed for derived imaging measures to derive harmonized outcomes. For voxel level analysis some standardization can be employed, such as RAVEL, but it is generally variable. Reach out to Biostats Core Dr. Tudorascu to discuss a harmonization plan if needed. Some useful references.(Torbati et al. 2021; Fortin et al. 2016, 2018) For IQ, there is a premorbid functioning level variable `prefunclevel`.

6.10 Clinical Lab and MOMs Data

Are clinical chemistry data (e.g., cholesterol) included?

Yes, this data is a standalone file on LONI

Are MOMs study data included?

Not yet, there will be a separate MOMs data file when the supplement ends in December 2024.

6.11 Gap

How was the gap between the visit and scanning/assay dealt with?

There are latency variables between all events

6.12 Errors

How are the data entry errors or sample switches handled?

LONI has a procedure

Is there a website so that data users know whether they are using the updated and correct data?

In the ABC-DS Project in LONI, the current data freeze is available; track the dates of your datasets; check the Ns;

6.13 Latency and Event Sequence

6.13.1 Event Sequence

Can you please confirm the variable `event_sequence == 1` refers to baseline and `clinical_latency_in_days` can be used to determine the time since baseline for all visits with `event_sequence > 1`.

Yes, confirmed.

6.13.2 Latency

What are the latency variables?

`clinical_latency_in_days`; `amy_latency_in_days`; `tau_latency_in_days`; `fdg_latency_in_days`; `csf_latency_in_days`; all use Cycle 1/baseline as the reference point.

6.14 Biospecimens/Omics

6.14.1 Biospeciman Dataset

How is the event sequence determined in the biospecimen dataset? Can we use `rundate` to determine the baseline visit? It seems that `rundate` may not be valid since it does not reflect the time of sample collection. In this case, how do we then determine baseline, and the relative timing (in days, months) of subsequent assessments?

The “Biospecimens” file should not be used, it was deleted in late August 2024, Use the CSF file.

6.14.2 Proteomics

Where is the proteomic file?

It is not on LONI yet. As of October 2024 we are waiting for the Methods document before the data can be posted.

References

- Ball, Sarah L, Anthony J Holland, Peter Treppner, Peter C Watson, and Felicia A Huppert. 2008. "Executive Dysfunction and Its Association with Personality and Behaviour Changes in the Development of Alzheimer's Disease in Adults with down Syndrome and Mild to Moderate Learning Disabilities." *British Journal of Clinical Psychology* 47 (1): 1–29.
- Beery, Keith E. 2004. "Beery VMI: The Beery-Buktenica Developmental Test of Visual-Motor Integration." *Minneapolis, MN: Pearson*.
- Chang, Christopher C., Carson C. Chow, Laurent C. A. M. Tellier, Sriram Vattikuti, Shaun M. Purcell, and James J. Lee. 2015. "Second-Generation PLINK: Rising to the Challenge of Larger and Richer Datasets." *GigaScience* 4: 7. <https://doi.org/10.1186/s13742-015-0047-8>.
- Chen, Zhicheng, David Mengel, Ashvini Keshavan, et al. 2019. "Learnings about the Complexity of Extracellular Tau Aid Development of a Blood-Based Screen for Alzheimer's Disease." *Alzheimer's & Dementia : The Journal of the Alzheimer's Association* 15 (3): 487–96. <https://doi.org/10.1016/j.jalz.2018.09.010>.
- Chhatwal, Jasmeer P, Aaron P Schultz, Yifan Dang, et al. 2020. "Plasma n-Terminal Tau Fragment Levels Predict Future Cognitive Decline and Neurodegeneration in Healthy Elderly Individuals." *Nature Communications* 11 (1): 6024. <https://doi.org/10.1038/s41467-020-19543-w>.
- Cummings, Jeffrey L, Michael Mega, Katherine Gray, Susan Rosenberg-Thompson, Daniela Anne Carusi, and Jeffrey Gornbein. 1994. "The Neuropsychiatric Inventory: Comprehensive Assessment of Psychopathology in Dementia." *Neurology* 44 (12): 2308–8.
- Devenny, DA, EJ Zimmerli, P Kittler, and SJ Krinsky-McHale. 2002. "Cued Recall in Early-Stage Dementia in Adults with down's Syndrome." *Journal of Intellectual Disability Research* 46 (6): 472–83.
- Esralew, Lucille, Matthew P Janicki, and Seth M Keller. 2017. "National Task Group Early Detection Screen for Dementia (NTG-EDSD)." In *Neuropsychological Assessments of Dementia in down Syndrome and Intellectual Disabilities*. Springer.
- Eurlings, H, H Evenhuis, and M Kenegen. 2006. "Dementia Questionnaire for People with

Learning Disabilities (DLD).” *UK Adaptation*.

Fortin, Jean-Philippe, Nicholas Cullen, Yvette I Sheline, et al. 2018. “Harmonization of Cortical Thickness Measurements Across Scanners and Sites.” *Neuroimage* 167: 104–20.

Fortin, Jean-Philippe, Elizabeth M Sweeney, John Muschelli, et al. 2016. “Removing Inter-Subject Technical Variability in Magnetic Resonance Imaging Studies.” *NeuroImage* 132: 198–212.

Galvin, JE, CM Roe, KK Powlishta, et al. 2005. “The AD8: A Brief Informant Interview to Detect Dementia.” *Neurology* 65 (4): 559–64.

Haxby, James V. 1989. “Neuropsychological Evaluation of Adults with down’s Syndrome: Patterns of Selective Impairment in Non-Demented Old Adults.” *Journal of Intellectual Disability Research* 33 (3): 193–210.

Illumina, Inc. 2020a. *GenomeStudio*. <https://www.illumina.com/techniques/microarrays/array-data-analysis-experimental-design/genomestudio.html>.

Illumina, Inc. 2020b. *GenomeStudio 2.0 Plug-in*. <https://support.illumina.com/downloads/genomestudio-2-0-plug-ins.html>.

Kaufman, Alan S. 2004. “Kaufman Brief Intelligence Test–Second Edition (KBIT-2).” *Circle Pines, MN: American Guidance Service*.

Korkman, Marit, Ursula Kirk, and Sally Kemp. 2007. *Nepsy-(nepsy-Ii)*. Harcourt Assessment San Antonio, TX.

Krinsky-McHale, Sharon J, Darlynnne A Devenny, Phyllis Kittler, and Wayne Silverman. 2008. “Selective Attention Deficits Associated with Mild Cognitive Impairment and Early Stage Alzheimer’s Disease in Adults with down Syndrome.” *American Journal on Mental Retardation* 113 (5): 369–86.

Manichaikul, Ani, Josyf C. Mychaleckyj, Stephen S. Rich, Kathy Daly, Michèle Sale, and Wei-Min Chen. 2010. “Robust Relationship Inference in Genome-Wide Association Studies.” *Bioinformatics* 26 (22): 2867–73. <https://doi.org/10.1093/bioinformatics/btq559>.

McCarthy, Dorothea. 1972. *MSCA: McCarthy Scales of Children’s Abilities*. Psychological Corporation.

Mengel, David, Shorena Janelidze, Robert J Glynn, Wen Liu, Oskar Hansson, and Dominic M Walsh. 2020. “Plasma NT1 Tau Is a Specific and Early Marker of Alzheimer’s Disease.” *Annals of Neurology* 88 (5): 878–92. <https://doi.org/10.1002/ana.25885>.

- Mengel, David, Wen Liu, Robert J Glynn, et al. 2020. "Dynamics of Plasma Biomarkers in down Syndrome: The Relative Levels of A 42 Decrease with Age, Whereas NT1 Tau and NfL Increase." *Alzheimer's Research & Therapy* 12 (1): 27. <https://doi.org/10.1186/s13195-020-00593-7>.
- Nasreddine, Ziad S, Natalie A Phillips, Valérie Bédirian, et al. 2005. "The Montreal Cognitive Assessment, MoCA: A Brief Screening Tool for Mild Cognitive Impairment." *Journal of the American Geriatrics Society* 53 (4): 695–99.
- Petersen, Melissa E, Fan Zhang, Nicole Schupf, et al. 2020. "Proteomic Profiles for Alzheimer's Disease and Mild Cognitive Impairment Among Adults with down Syndrome Spanning Serum and Plasma: An Alzheimer's Biomarker Consortium-down Syndrome (ABC-DS) Study." *Alzheimer's & Dementia (Amsterdam, Netherlands)* 12 (1): e12039. <https://doi.org/10.1002/dad2.12039>.
- Petersen, Melissa, Fan Zhang, Sharon J Krinsky-McHale, et al. 2020. "Proteomic Profiles of Prevalent Mild Cognitive Impairment and Alzheimer's Disease Among Adults with down Syndrome." *Alzheimer's & Dementia (Amsterdam, Netherlands)* 12 (1): e12023. <https://doi.org/10.1002/dad2.12023>.
- Reiss, Steven. 1994. *Handbook of Challenging Behavior: Mental Health Aspects of Mental Retardation*. Ids Pub.
- Sparrow, SS, CA Saulnier, and DC Cicchetti. 2016. *Vineland Adaptive Behavior Scales (3rd Ed.)*. Bloomington, MN: Pearson.
- Tiffin, Joseph, and Eston Jackson Asher. 1948. "The Purdue Pegboard: Norms and Studies of Reliability and Validity." *Journal of Applied Psychology* 32 (3): 234.
- Tinetti, Mary E. 1986. "Performance-Oriented Assessment of Mobility Problems in Elderly Patients." *Journal of the American Geriatrics Society*.
- Torbati, Mahbaneh Eshaghzadeh, Davneet S Minhas, Ghasan Ahmad, et al. 2021. "A Multi-Scanner Neuroimaging Data Harmonization Using RAVEL and ComBat." *Neuroimage* 245: 118703.
- Wechsler, D. 2003. *WISC-IV. Wechsler Intelligence Scale for Children—Fourth UK Edition. Administration and Scoring Manual*. Harcourt Assessment, London.(tr. it.: WISC-IV. Wechsler Intelligence Scale